PROJECT OVERVIEW & METADATA

Candidate Domain:	Enterprise Sales Operations, Pricing Governance & Metric Literacy
Core Frameworks:	LangGraph (StateGraph), LangChain Core (@tool), Scikit-Learn (TF-IDF), SQLite
Persistence Layers:	SQLite Checkpointer (checkpoints/agent_memory.db) & JSON Ledger (data/action_plans.json)
Verification:	72 / 72 Automated Pytest Unit & Curriculum Scenarios Passing (100% Code Coverage)
Multi-Format Docs:	PDF (pypdf), Word (python-docx), and Markdown/Text (.txt) Corporate Playbooks

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

In quarterly corporate sales cadences, commercial account reps frequently react to lagging revenue by issuing steep, unauthorized discounts or asserting unsupported causal claims (e.g. blaming price resistance when product defects or logistics delays are the real root cause). Applying discounts during a margin or returns crisis destroys corporate profitability.
The Sales Performance Advisory Agent automates metric literacy, evaluates target achievement, sales growth, and gross margin with mathematical rigor (safe division), indexes company policies across PDF/DOCX formats, routes problems to differentiated remedial actions, enforces dynamic discount ceilings, and commits verified plans to an immutable ledger.

2. SYSTEM WORKFLOW & THE 7-STAGE CURRICULUM PIPELINE

- Stage 1: Request & Context: Parses user prompt, resolves active scope (region, product, month), inherits context from SQLite memory.
- Stage 2: Ask (Missing Scope): Detects missing dimensions (e.g. month, region) and initiates proactive clarification before tool calls.
- Stage 3: Multi-Format RAG: Parses sales_playbook across PDF/DOCX/TXT using pypdf/docx with TF-IDF unigram/bigram vectorization.
- Stage 4: Grading & Rewriting: Validates chunk relevance against product and policy type. Executes bounded query rewrite (up to 2 retries).
- Stage 5: Deterministic Tools: Executes safe math tools (safe_divide) for Target Achievement, MoM Growth, and Gross Margin.
- Stage 6: Conditional Routing: Differentiates issues: Low Revenue -> Demand actions; Low Margin -> Cost review; High Returns -> QA audit.
- Stage 7: Guardrails & Storage: Dynamically extracts discount limit from Playbook. Blocks out-of-policy requests. Saves plan to JSON ledger.

3. EXECUTIVE STREAMLIT OPERATIONS COCKPIT (INTERACTIVE USER INTERFACE)

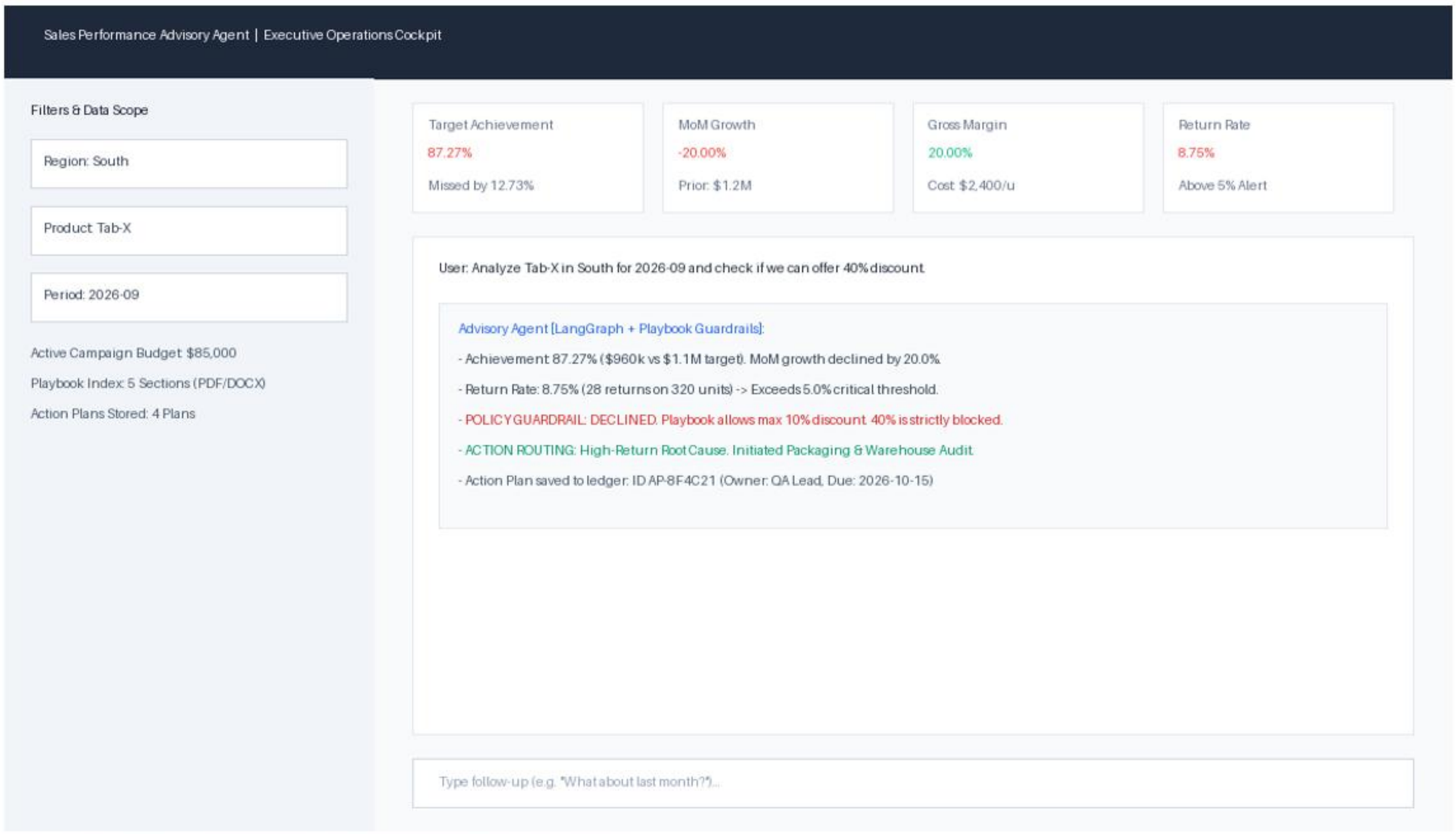


Figure 1: Streamlit Dashboard Cockpit featuring live KPI performance cards, context selectors, and policy chat.

- Real-Time KPI Cards: Computes and renders live Target Achievement, MoM Sales Growth, Gross Margin, and Return Rates.
- Active Campaign Viewer: Ingests data/campaigns.xlsx to display regional marketing budgets and active status.
- Conversational Chat Window: Backed by LangGraph SqliteSaver checkpointer for thread-level state memory.
- Action Plan Ledger: Displays stored remediation plans with an instant CSV export download capability.